Desk Research Process Mapping for Theoretical Models

Introduction and Research Context

This study progressed from methodological design into active data collection within the Value Chain Hackers Lab at Windesheim University of Applied Sciences. The lab operates as an applied research environment in which student teams collaborate with external organisations on challenges related to transparency, sustainability, and system resilience. Projects are conducted within fixed academic timelines and shaped by practical constraints, including variation in student research experience, stakeholder availability, and the operational realities of working with real organisations in real time.

The rationale for this research is grounded in the principle that the credibility of applied research depends upon transparency, traceability, and reproducibility. When projects extend across student cohorts, weaknesses in documentation and continuity can lead to repeated loss of research value. This generates micro level consequences for students who begin new projects without usable records of prior work, meso level consequences for supervisors who cannot reliably compare projects or accumulate insight over time, and macro level consequences as research practices risk misalignment with broader expectations for open, verifiable, and reusable scholarship. These expectations are reflected in international policy and governance frameworks that emphasise findability, accessibility, interoperability, and reuse, alongside strengthened stewardship of research outputs (Wilkinson et al., 2016; OECD, 2021; Science Europe, 2021; European Commission, n.d.).

Because this study is conducted within a functioning research laboratory rather than a controlled experimental setting, it unfolds within an uncontrolled real world research context. This condition does not diminish the requirement for rigor; rather, it alters how rigor must be achieved. Within applied case study research, rigor is preserved by maintaining a traceable chain of research logic, explicitly explaining methodological decisions, and demonstrating how evidence informs both method selection and analytical conclusions (Yin, 2018). When elements of an initial design cannot be executed as anticipated, the researcher must implement controlled adjustments, justify those decisions transparently, and demonstrate that such adaptations protect data quality rather than weaken it (Yin, 2018).

The Graduation Project Plan established the initial methodological direction through a qualitative design integrating desk research with field research. The overall objective is to design and validate a reproducible research framework capable of strengthening transparency, traceability, and reuse within student led applied research projects. This orientation aligns with design science principles, in which a framework must be grounded in a real and observable problem, developed using established knowledge, and evaluated through a structured research cycle (Hevner et al., 2004; Peffers et al., 2007).

A central principle guiding the study is that research logic must remain ordered and auditable through a clearly articulated progression from the IS situation to the SOLL situation. The IS situation refers to the evidenced current state of research practice within the VCH Lab, representing the observable reality of how research activities were executed, documented, and transferred at the time of investigation. The SOLL situation represents the justified target state describing the future condition that research practices should achieve in order to meet higher standards of transparency, reproducibility, methodological rigor, and academic defensibility. The analytical gap between these states forms the foundation for defining design requirements and later validating the proposed framework.

Maintaining this ordering is essential for methodological independence. Field research first establishes the IS situation within its operational context. Desk research then identifies theoretical requirements and quality standards capable of explaining why the current state produces loss of traceability and weakened continuity, while clarifying the conditions necessary to prevent recurrence. Operationalisation subsequently translates these requirements into examinable constructs. Design science principles guide how the framework is developed and evaluated, while lifecycle reference models help determine when traceability activities should occur throughout the research process rather than being deferred to end stage documentation. Within this study, EDRM was used analytically as a lifecycle reference model to structure when traceability activities should occur across the research process, rather than as a theoretical or methodological foundation (EDRM, n.d.). This sequencing ensures that the target state emerges from empirical evidence and validated standards rather than conceptual preference.

A framework oriented approach was selected because commonly proposed improvements within educational environments often address surface consistency without resolving the underlying traceability problem. Increased oversight may strengthen short term alignment but does not scale effectively and may constrain student autonomy, a critical condition for meaningful learning (Biggs & Tang, 2011). Templates and checklists can provide structural guidance, yet they do not inherently connect data, reasoning, and conclusions in a manner that supports independent verification and reuse (Hevner et al., 2004; OECD, 2021). For this reason, the study focuses on establishing a reproducible research flow rather than merely improving the format of final reports.

This positioning reflects the conditions under which applied research must maintain methodological transparency while operating within authentic organisational settings

Identification of the IS Situation Through Field Research

The study began by establishing the IS situation within the laboratory environment. In this context, the IS situation denotes the current observable condition of research practice, including how activities were conducted, how decisions were recorded, how documentation was produced, and how knowledge was stored and transferred across cohorts. Defining this starting point through empirical evidence ensures that subsequent improvements are grounded in verified conditions rather than assumption.

Field research drew upon complementary sources of evidence. Document analysis examined completed project records and final deliverables, focusing on the visibility of key decisions, intermediate analytical steps, source usage, and validation logic. Fully structured interviews were conducted with individuals who held direct responsibility for executing or supervising research projects. This approach aligns with purposive sampling logic, whereby participants are selected based on direct exposure to the practices under investigation, thereby supporting depth of insight and strong alignment with the research question (Patton, 2015).

The field research strategy further relied upon triangulation. Document analysis reveals what has been preserved within project materials, while interviews clarify how research work unfolded in practice, including gaps not immediately visible within documented records. Examining patterns across both forms of evidence strengthens interpretive credibility by preventing reliance on a single data source (Yin, 2018).

Emergent IS Situation

The field research revealed a consistent structural pattern. Although project outcomes frequently retained practical value, the analytical pathways through which conclusions were reached were not always reconstructable. Decision rationales, intermediate analyses, and supporting documentation were at times fragmented, inconsistently stored, or absent. Consequently, later cohorts often depended on verbal explanation to understand prior work. When continuity relies primarily on social memory rather than durable research records, verification becomes more difficult and reuse potential declines. This observation aligns with reproducibility scholarship demonstrating that limited workflow transparency weakens cumulative knowledge development (Peng, 2011).

This pattern signals a structural vulnerability rather than an individual performance issue. The central concern is not whether students demonstrated effort, but whether the research process consistently generates traceable records that enable an independent reviewer to reconstruct decisions, data lineage, and analytical reasoning. Provenance was therefore treated as a foundational concept early in the study. Provenance refers to information describing how a result was produced, including the entities involved, the activities performed, and the actors responsible, so that the outcome can be examined and reconstructed (W3C, 2013).

The recurrence of these conditions across projects further supports a structural interpretation. Socio technical theory explains that organisational breakdown typically emerges from the interaction between routines, tools, roles, and human behaviour rather than from isolated personal shortcomings (Trist, 1981; Mumford, 2006). The IS situation was therefore defined as a structural weakness in research traceability and continuity within an applied laboratory environment.

Institutional and System-Level Benchmarking of Research Documentation Practices

The structural conditions observed within the VCH Lab are not unique to this environment. Across the Dutch research ecosystem, documentation is increasingly treated as a governed component of responsible research practice rather than an individual preference. Examining comparable governance structures therefore provides contextual validation that the diagnosed IS situation reflects a recognised research continuity challenge rather than a purely local operational issue. Collectively, these environments indicate that documentation maturity emerges from embedded research infrastructure rather than individual researcher behaviour.

At the national level, the Netherlands Organisation for Scientific Research requires researchers to address research data management during funding applications and to develop formal data management plans following grant approval. These expectations compel early decisions regarding storage, stewardship, sharing conditions, and long term preservation, positioning documentation as a structural research control that safeguards the cumulative value of scientific work (NWO, n.d.). This national framing is directly relevant to the VCH context, where the primary risk is not the absence of project outputs but the inability of subsequent cohorts to reconstruct the analytical pathway from data to conclusion.

At the institutional level, universities increasingly operationalise these expectations through formal research data management policies. The University of Twente explicitly commits to responsible and transparent research data management while aligning institutional practice with FAIR oriented principles and clearly defined role responsibilities (University of Twente, n.d.). By standardising expectations across projects, such frameworks reduce variability in documentation quality and enable supervisors to compare research efforts more reliably, directly addressing a key meso level vulnerability identified within the IS situation.

National infrastructure providers further reinforce lifecycle oriented documentation practices by supporting research environments that extend beyond active storage toward structured archiving and publication. SURF supports platforms such as Yoda that preserve research materials across the full data lifecycle. Crucially, structured metadata is required before research packages can be archived, ensuring that contextual information necessary for interpretation is captured as part of workflow rather than appended retrospectively (SURF, n.d.). This approach directly addresses failures observed in the IS diagnosis, where fragmented storage and unclear relationships between files, decisions, and outputs limited reconstructability.

Operational verification practices demonstrate an additional sector shift from preservation toward reproducibility assurance. TU Delft’s Digital Competence Centre supports services that evaluate whether workflows and research processes can be executed to reproduce reported outputs, reinforcing the expectation that research should remain independently verifiable (TU Delft Digital Competence Centre, n.d.). Similarly, the 4TU.ResearchData consortium provides infrastructure for publishing and preserving research outputs with persistent identifiers, supporting long term findability, reuse, and validation while reducing reliance on informal knowledge transfer (4TU.ResearchData, n.d.).

At the workflow level, collaborative research communities such as The Alan Turing Institute emphasise embedding reproducibility directly within the working research environment. Through resources such as *The Turing Way*, documentation is integrated into daily workflow via standardised project structures that support the continuous recording of methodological decisions, source usage, and analytical processes (The Alan Turing Institute, n.d.). Embedding traceability within the project workspace reduces dependence on retrospective reconstruction and strengthens procedural transparency throughout the research lifecycle.

Taken together, these national, institutional, infrastructural, operational, and workflow level approaches reveal a consistent principle: research continuity is sustained when documentation is structured, embedded within routine work, supported by governance expectations, and reinforced through verification practices. This convergence strengthens the interpretation of the IS situation as a structural condition rather than an isolated operational weakness and provides a credible external benchmark for developing a reproducible research framework capable of supporting traceable and reusable research within an applied laboratory environment.

Transition from GP1 Expectations to the Executed Research Process

The methodological expectations outlined in GP1 included surveys as an additional source of insight alongside document analysis and interviews. During execution, this component required adjustment due to practical timing constraints. Survey distribution coincided with the academic holiday period, and by the time active data collection began, the opportunity for meaningful participation had diminished. Continuing with delayed distribution risked collecting responses detached from active project work, thereby weakening procedural accuracy.

The study therefore prioritised methods capable of capturing both procedural depth and system level insight. The executed approach relied on fully structured interviews with former project leads from recently completed case studies, complemented by interviews with lab supervisors responsible for continuity and research expectations across cohorts. This adjustment did not alter the intent of the research design; rather, it preserved the objective of diagnosing enacted research practices while selecting a feasible strategy that protected data relevance within an uncontrolled setting (Yin, 2018).

The interview approach was deliberately constructed rather than improvised. A fully structured format ensured comparability across participants and reduced variation in question delivery, strengthening interpretability when examining recurring patterns in research practice. The interview guide was developed through a controlled design process that translated analytical constructs into fixed question wording and sequencing, thereby supporting methodological clarity and auditability (Kallio et al., 2016; Patton, 2015).

Analytical Trigger for Desk Research

Field research did not function as an endpoint but as an analytical trigger. Once the IS situation had been defined as a structural weakness in traceability and continuity, the next step was to identify the mechanisms capable of explaining why these breakdowns occur and what conditions must be established to prevent them.

Desk research was therefore undertaken to locate theoretical models and reproducibility standards able to explain the observed conditions and translate them into defensible constructs and requirements for framework design and evaluation. This sequencing is critical for methodological legitimacy. Theoretical inquiry did not invent the problem; it explained an empirically established condition and guided the subsequent operationalisation of the research design (Hevner et al., 2004; Wilkinson et al., 2016; OECD, 2021; Science Europe, 2021).

Research Strategy and Search Orientation

The desk research followed a staged strategy designed to balance conceptual breadth with methodological control. The first stage focused on broad problem grounding by identifying models capable of explaining reproducibility breakdown, traceability loss, documentation failure, and weak continuity within applied research environments. The objective was to confirm that the diagnosed condition possesses defensible theoretical substance rather than representing an isolated operational concern.

The second stage retained the same core framing while deliberately shifting the analytical lens toward educational context, organisational learning, and knowledge continuity. This variation supported convergence testing. When independent searches reveal similar mechanisms and priorities, confidence increases that the theoretical foundation reflects stable explanatory dynamics rather than a single interpretive perspective.

The third stage marked a transition from exploration to refinement. Models were evaluated against the observed environment and the practical requirements of framework design, validation, and long term sustainability. Inclusion was guided by whether a model strengthened problem diagnosis, reinforced research design legitimacy, clarified evaluation logic, or supported continuity across cohorts. This disciplined selection prevented uncontrolled theoretical expansion and ensured that the foundation remained purposeful, coherent, and defensible (Hevner et al., 2004; Peffers et al., 2007; Yin, 2018).

Across all stages, the governing standard was that methodological and design choices must possess a traceable intellectual origin supported through APA cited evidence. Field findings, validated constructs, theoretical models, and methodological scholarship constitute acceptable foundations for academic reasoning, whereas intuition alone does not provide sufficient justification within rigorous research (Yin, 2018; Hevner et al., 2004).

Section 2: Stage One

Broad Theoretical Grounding

Purpose and Logic of Stage One

Stage One commenced once the IS situation had been empirically established as a structural weakness in research traceability and continuity within the Value Chain Hackers Lab. At this point, the study required more than descriptive clarity. It required theoretical explanation. The purpose of Stage One was therefore to determine whether the observed condition possesses recognised academic substance and can be explained through established models, rather than being interpreted as a local operational irregularity. **This interpretation is further supported by system-level and institutional benchmarking presented earlier, which demonstrated that mature research environments formalise documentation precisely to safeguard continuity, traceability, and reuse.**

This progression aligns directly with design science reasoning. Before a reproducible research framework can be responsibly developed, the problem it addresses must be grounded in a clear understanding of what is occurring within the real environment and why those conditions persist (Hevner et al., 2004; Peffers et al., 2007). Stage One was therefore intentionally broad yet analytically controlled, not to accumulate theory, but to identify explanatory perspectives capable of illuminating the mechanisms behind reproducibility breakdown and to test whether those perspectives align with the conditions observed within the VCH Lab.

Stage One also functioned as a methodological safeguard. Framework-oriented research cannot be justified solely as a practical improvement initiative. It must be positioned as a response to known and studied dynamics, such as fragmented workflow visibility, weak provenance, socio-technical misalignment, and disrupted continuity across project cycles (Peng, 2011; Trist, 1981; Mumford, 2006). Establishing this grounding ensures that the emerging SOLL situation is derived from validated knowledge rather than conceptual preference.

Importantly, this theoretical grounding proceeds from a position already strengthened by macro contextual validation. Institutional and infrastructural comparators demonstrated that documentation governance is increasingly treated as a prerequisite for credible research rather than an administrative afterthought. Stage One therefore advances with reinforced problem legitimacy: the diagnosed condition reflects a recognised research governance challenge rather than an isolated laboratory concern.

Search Framing and Selection Criteria

Stage One was guided by the study's main research question, which focuses on designing and validating a reproducible research framework capable of making student-led research transparent, verifiable, and reusable within an institutionally reliable structure. The theoretical search was deliberately broad enough to capture relevant explanatory traditions, yet sufficiently controlled to prevent the accumulation of theory without analytical purpose.

Models were considered relevant when they satisfied at least one of three conditions.

First, a model needed to explain why traceability and reuse tend to deteriorate within applied research environments.

Second, a model needed to strengthen the academic legitimacy of designing and evaluating a structured framework within a real-world context.

Third, a model needed to provide operational clarity regarding what transparency, traceability, and reuse require in practice, ensuring that later evaluation would not depend on subjective judgement.

Applying these criteria preserved methodological discipline and ensured that each theoretical inclusion could be justified as serving a defined analytical function. This approach reflects case study guidance emphasising the importance of maintaining a visible chain of reasoning so that readers can audit how conceptual decisions emerge from empirical observation (Yin, 2018). Maintaining this traceable reasoning is particularly important in applied research settings, where theoretical choices must remain visibly connected to observed organisational conditions.

Models Identified in Stage One and Their Analytical Contribution

The first model identified was **Work System Theory**, which conceptualises organisations as integrated systems in which participants perform activities using information and technology to produce outcomes for stakeholders (Alter, 2013). This perspective proved highly relevant because the VCH Lab operates not only as an educational setting but also as a functioning research environment characterised by recurring processes, defined roles, supporting tools, and structured outputs.

From a work system perspective, traceability breakdown can be interpreted as a condition of system misalignment. Unclear documentation responsibilities, inconsistent storage routines, fragile handover structures, and uneven tooling practices can collectively produce repeated loss of decision visibility. Viewing the IS situation through this lens strengthened the structural diagnosis by demonstrating that documentation failure is unlikely to be resolved through individual effort alone. Instead, it must be addressed through improved alignment between roles, workflows, and informational expectations (Alter, 2013).

The second model identified was **Design Science Research**, which is central to the methodological posture of this study. The research does not merely describe a problem; it seeks to develop and validate a reproducible research framework capable of improving practice. Design science establishes that scholarly knowledge can be generated through the creation and evaluation of a structured solution addressing a real-world challenge (Hevner et al., 2004).

This model clarified the conditions required for academic legitimacy. The framework must be grounded in an observable problem, informed by established knowledge, and evaluated using explicit criteria. Positioning the study within design science therefore ensures that the framework is interpreted as a research contribution rather than as operational guidance alone (Hevner et al., 2004; Peffers et al., 2007). It also reinforces methodological coherence by linking problem diagnosis directly to solution-oriented inquiry.

The third model identified concerned **provenance**, defined as structured information describing how a research outcome was produced, including the entities involved, the activities performed, and the responsible actors. Provenance directly determines whether an independent reviewer can reconstruct the pathway from data to conclusion (W3C, 2013).

This perspective was particularly relevant because field research indicated that projects often retained visible outcomes while the reasoning behind those outcomes remained partially obscured. Provenance models therefore clarified that traceability requires more than stored files. It requires explicit relational links between inputs, analytical actions, and resulting conclusions. This insight provided operational precision for later design requirements by specifying what reconstructability must contain.

The fourth model identified was the **FAIR guiding principles**, which define research quality in terms of findability, accessibility, interoperability, and reuse (Wilkinson et al., 2016). FAIR proved especially valuable because the continuity challenge observed within the VCH Lab is fundamentally a reuse problem. When prior research cannot be located, interpreted, or trusted, its cumulative value diminishes regardless of the effort invested in its production.

FAIR also introduced an externally recognised benchmark for the SOLL situation, allowing the study to position its target state within broader expectations for responsible research stewardship rather than relying solely on internal standards. This alignment strengthens macro credibility and reflects practices increasingly embedded within mature research infrastructures and governed data environments (OECD, 2021; Science Europe, 2021).

The fifth model identified was **Normalization Process Theory**, which explains how new practices become embedded within routine work through processes of shared understanding, engagement, enactment, and ongoing appraisal (May & Finch, 2009). This perspective was critical because the effectiveness of a reproducible research framework depends not only on its technical structure but also on its capacity to become integrated into everyday research behaviour.

Within an environment characterised by cohort turnover, practices that remain optional or externally imposed rarely persist. Normalization Process Theory therefore strengthened the long-term continuity logic by demonstrating that sustainable improvement requires practices to become perceived as intrinsic to competent research rather than as administrative obligations.

Stage One Outcome

Stage One confirmed that the IS situation can be explained through established theory and that the diagnosed condition is structurally meaningful rather than incidental. Work System Theory supported the interpretation of documentation breakdown as a system alignment challenge rather than a purely individual shortcoming (Alter, 2013). Design science established the academic legitimacy of developing and evaluating a reproducible research framework as a structured response to the observed condition (Hevner et al., 2004; Peffers et al., 2007). Provenance clarified the informational requirements necessary for reconstructable research pathways (W3C, 2013). FAIR provided an externally recognised benchmark for reuse and long-term research value (Wilkinson et al., 2016). Normalization Process Theory strengthened the sustainability argument by demonstrating that continuity depends on practices becoming embedded within routine workflows (May & Finch, 2009).

The convergence of these models indicates that the diagnosed condition is supported by multiple explanatory traditions rather than a single interpretive lens, thereby strengthening theoretical sufficiency and reducing the risk of conceptual fragility as the research progresses.

Together, these models provided sufficient theoretical grounding to proceed beyond descriptive diagnosis. The study was therefore positioned to enter the next stage of inquiry, where the objective shifts from locating explanatory theory toward testing the stability of that explanation through independent expansion and convergence. This progression preserves analytical discipline and ensures that the theoretical foundation remains both purposeful and defensible as the research advances toward defining the SOLL situation (Yin, 2018).

Section 3: Stage Two

Independent Theoretical Expansion and Convergence

Purpose and Logic of Stage Two

Stage Two was conducted to examine whether the theoretical explanation established in Stage One remained stable when the IS situation was analysed through a deliberately different conceptual orientation. This step is methodologically significant because a single theoretical pass can unintentionally reflect the perspective of the first body of literature encountered rather than the most structurally accurate explanation. Introducing a second analytical lens reduces this risk by testing whether alternative scholarly traditions identify similar underlying mechanisms.

The purpose of Stage Two was therefore convergence testing. If an independent exploration produces theoretical perspectives that continue to point toward the same structural conditions, confidence increases that the diagnosed problem is not dependent on one interpretive framing but instead reflects durable dynamics within the research environment. Within applied research, such convergence strengthens interpretive credibility by demonstrating that the explanation is supported across multiple domains of scholarship rather than anchored to a single theoretical tradition (Yin, 2018).

Where Stage One confirmed that the IS situation possesses recognised academic substance, Stage Two evaluated the stability of that substance. The analytical objective shifted from establishing theoretical legitimacy toward verifying that the emerging explanatory structure remains coherent under conceptual variation.

Importantly, Stage Two did not represent uncontrolled theoretical expansion. The search remained anchored to the same research objective: explaining reproducibility breakdown and identifying the conditions required to support traceable, reusable research across cohort transitions. This continuity ensured that theoretical inclusion remained governed by analytical necessity rather than intellectual curiosity.

Stage Two also increased sensitivity to the educational and organisational characteristics that define the VCH Lab environment. Field evidence indicated that cohort turnover, informal knowledge transfer, uneven documentation routines, and time-bound project cycles significantly influence research continuity. The theoretical lens therefore widened toward educational design, organisational learning, and knowledge conversion while remaining structurally aligned with the IS to SOLL progression.

This progression reflects disciplined research reasoning. Premature theoretical closure can produce fragile foundations, while controlled expansion enables the researcher to determine whether explanatory patterns persist across adjacent academic domains. The aim was not theoretical breadth, but theoretical resilience.

Search Framing and Disciplined Inclusion Logic

Stage Two retained the core search framing established earlier while adjusting the analytical emphasis. Rather than prioritising governance structures and traceability models alone, the search explored theories capable of explaining how knowledge is generated, stabilised, transferred, and sustained within learning-oriented applied environments.

Despite this broader orientation, inclusion criteria remained unchanged in principle. Models were retained only when they strengthened at least one of three analytical requirements:

• deepening explanation of why traceability and reuse deteriorate within applied student research environments
• reinforcing the legitimacy of designing and validating a reproducible research framework under real operational constraints
• providing operational insight capable of informing workflow structure, documentation practices, or continuity mechanisms

Maintaining these criteria prevented theoretical drift and preserved the chain of reasoning linking conceptual choices to the empirically established IS situation (Hevner et al., 2004; Yin, 2018). Each retained perspective therefore performs a defined analytical role rather than contributing conceptual density without purpose.

This disciplined approach is particularly important in applied research contexts, where excessive theoretical breadth can weaken methodological clarity. By ensuring that every inclusion remained visibly connected to the diagnosed condition, the study preserved interpretive transparency while strengthening theoretical robustness.

Models Identified in Stage Two and Their Analytical Contribution

The first model identified was **Design-Based Research**, an iterative methodology centred on developing and refining interventions within real-world learning environments through cycles of design, enactment, analysis, and revision (Wang & Hannafin, 2005; Anderson & Shattuck, 2012).

This perspective proved highly relevant because the VCH Lab operates simultaneously as a research environment and an educational setting. Students are expected to deliver applied outcomes while still developing research competence. Design-Based Research therefore strengthened contextual legitimacy by demonstrating that iterative refinement can remain academically rigorous when it is systematically documented, theoretically informed, and analytically evaluated.

Importantly, this model complements rather than replaces Design Science Research. Design science provides the primary methodological legitimacy for producing a reproducible research framework as a scholarly contribution (Hevner et al., 2004; Peffers et al., 2007). Design-Based Research enhances contextual fit by explaining how structured iteration can be responsibly conducted in environments where learning is intrinsic to the research process. Together, these perspectives reinforce the claim that movement from the IS situation toward the SOLL situation must occur through controlled and traceable cycles rather than informal adjustment.

The second model concerned **socio-technical systems theory** viewed through a broader organisational learning lens. Socio-technical theory posits that outcomes depend on the joint optimisation of social and technical subsystems. Tools, workflows, incentives, role expectations, and cultural norms must operate in alignment, or system performance becomes unstable (Trist, 1981; Mumford, 2006).

This remained directly relevant because the traceability breakdown observed within the VCH Lab cannot be explained solely as incomplete documentation. It reflects interactional misalignment between supervision routines, student incentives, time constraints, tooling structures, and inconsistent documentation expectations. Stage Two therefore strengthened confidence that the IS situation represents a systemic condition rather than a narrow behavioural habit.

This insight carries immediate implications for the SOLL orientation. A reproducible research framework must address behavioural and structural alignment rather than focusing exclusively on technical storage solutions. Without socio-technical coherence, even well-designed documentation structures risk partial adoption.

The third model identified was **knowledge management theory**, particularly the SECI model, which explains how knowledge transitions between tacit and explicit forms through processes of socialisation, externalisation, combination, and internalisation (Nonaka & Takeuchi, 1995; Nonaka & von Krogh, 2009).

This perspective proved especially illuminating because field evidence indicated that later cohorts often relied on verbal explanation to understand prior analytical work. Such reliance strongly suggests that critical reasoning remains tacit rather than being externalised into durable research records.

Interpreting the IS situation through SECI reframed traceability breakdown as a knowledge conversion failure. When analytical reasoning is not externalised, it becomes vulnerable to loss during team transition, producing recurring discontinuity even when visible outputs appear complete. The model therefore supports a clear SOLL implication: the framework must actively facilitate structured externalisation so that analytical pathways remain reconstructable across cohorts.

The fourth model identified was **open science lifecycle thinking**, including structures associated with collaborative research environments such as the Open Science Framework. Open science perspectives emphasise that transparency must be embedded throughout the research lifecycle, spanning planning, methodological articulation, analytical visibility, and long-term preservation (Nosek et al., 2015; Fecher & Friesike, 2014).

This orientation was highly relevant because traceability breakdown is frequently temporal. It emerges gradually as projects evolve and intermediate decisions become obscured. Lifecycle models therefore strengthened the argument that reproducibility must be structurally integrated across phases of work rather than appended during final reporting.

Within the IS to SOLL progression, this represents a shift from reactive reconstruction toward proactive research design. Traceability becomes an ongoing research behaviour rather than a retrospective administrative exercise.

The fifth observation was the continued recurrence of the **FAIR guiding principles** despite the shift in analytical lens. FAIR remained relevant even when the search expanded toward educational design and knowledge continuity, reaffirming that research outputs must be findable, accessible, interoperable, and reusable, supported by sufficient contextual metadata (Wilkinson et al., 2016; Mons et al., 2017).

The independent reappearance of FAIR is analytically significant. When a framework persists across multiple theoretical explorations, it signals conceptual stability rather than selective interpretation. FAIR therefore strengthened its position as a macro-level SOLL benchmark, aligning the proposed framework not only with internal continuity needs but also with broader expectations for responsible research stewardship (OECD, 2021; Science Europe, 2021).

Convergence Analysis and What Stage Two Demonstrated

The primary contribution of Stage Two lies not in the accumulation of additional theory but in confirming the recurrence of the same structural mechanisms across distinct scholarly traditions.

Stage One highlighted system alignment, provenance requirements, reuse standards, and the sustainability of practice. Stage Two, despite applying a different conceptual orientation, pointed toward identical foundational conditions:

Traceability depends on socio-technical alignment.
Reuse depends on explicit standards and metadata.
Continuity depends on converting tacit reasoning into durable records.
Sustainability depends on practices becoming embedded within routine workflows.

This convergence is methodologically consequential. Explanations supported by multiple theoretical lenses are less vulnerable to critique that theory was selected to fit the project. Instead, the diagnosed condition appears structurally grounded and resilient to conceptual variation.

Within applied research environments, such resilience strengthens interpretive credibility and provides a more secure basis for deriving design requirements that define the SOLL situation (Yin, 2018).

Stage Two Outcome

Stage Two confirmed the stability of the theoretical explanation while improving contextual fit. Design-Based Research strengthened legitimacy for iterative improvement within an educational applied environment (Anderson & Shattuck, 2012; Wang & Hannafin, 2005). Socio-technical systems theory reinforced that reproducibility breakdown is shaped by the interaction of social and technical routines (Trist, 1981; Mumford, 2006). The SECI model clarified that traceability failure is simultaneously a knowledge conversion failure (Nonaka & Takeuchi, 1995; Nonaka & von Krogh, 2009). Open science lifecycle perspectives reinforced the necessity of embedding transparency across the research process (Nosek et al., 2015; Fecher & Friesike, 2014). The repeated relevance of FAIR confirmed the stability of reuse requirements across theoretical perspectives (Wilkinson et al., 2016; Mons et al., 2017).

At this point, the research no longer depended on exploratory theory identification. It had achieved theoretical stability.

The study was therefore positioned to proceed toward disciplined prioritisation, where the objective shifts from confirming explanatory validity to selecting the models most capable of guiding framework design, defining evaluation criteria, and supporting long-term continuity under conditions of cohort transition.

Stage Three thus advances from a position of conceptual confidence rather than theoretical uncertainty, enabling the research to move from convergence toward purposeful architectural alignment within the IS to SOLL progression.

Section 4: Stage Three

Theoretical Prioritisation and Contextual Alignment

Purpose and Logic of Stage Three

By the conclusion of Stage Two, the study had established a theoretically stable foundation supported by convergence across multiple scholarly traditions. At this point, further expansion of the theoretical landscape would no longer strengthen the research. Instead, it would introduce the risk of conceptual diffusion, whereby excessive breadth weakens methodological clarity rather than improving it.

The purpose of Stage Three was therefore prioritisation. The research shifted from theoretical expansion toward disciplined selection, ensuring that the final model set directly supports four core analytical requirements:

- diagnosing the structural mechanisms underlying the IS situation
- legitimising the research design
- guiding the development of a reproducible research framework
- defining evaluative criteria aligned with the intended SOLL situation

This transition reflects established case study guidance emphasising the importance of maintaining a clear chain of evidence so that readers can trace how conceptual commitments emerge from empirical observation and analytical reasoning (Yin, 2018). It also aligns with design science principles that stress purposeful knowledge utilisation when constructing a solution intended to address a real-world problem (Hevner et al., 2004).

Within the IS to SOLL progression, Stage Three functions as a stabilisation point in the theoretical architecture. The IS situation established what is occurring and why structural vulnerabilities persist. Stage Two confirmed that these vulnerabilities are supported by convergent theory. Stage Three determines which perspectives possess sufficient explanatory depth and operational relevance to guide the transition toward a defensible future state.

This movement from expansion to prioritisation signals methodological maturity. Rather than demonstrating familiarity with a wide theoretical field, the research now demonstrates the capacity to exercise analytical judgement regarding which knowledge is necessary.

Evaluation Criteria for Theoretical Retention

Models were retained only when they satisfied at least one of the following conditions:

- clarifying the structural mechanisms producing traceability breakdown
- strengthening the academic legitimacy of framework-oriented research
- providing operational guidance capable of informing workflow structure or lifecycle integration
- defining measurable conditions for reuse, continuity, and institutional reliability

Applying these criteria ensured methodological control and prevented the emerging framework from appearing theoretically dense without analytical purpose. Each retained model therefore performs a defined research function within the progression from diagnosis toward design.

Such restraint is particularly important in applied research environments, where theoretical overextension can obscure rather than support practical contribution. By limiting the final model set to perspectives that directly inform design and evaluation, the study preserved conceptual coherence while strengthening analytical defensibility.

Models Prioritised in Stage Three and Their Contribution

Open Science Lifecycle Logic

Open science lifecycle perspectives were prioritised because they translate transparency from an abstract research value into structured behaviour across project phases. Contemporary research credibility increasingly depends on the visibility of planning decisions, methodological pathways, analytical processes, and preserved outputs (Nosek et al., 2015; Fecher & Friesike, 2014).

This orientation proved directly relevant to the IS situation, where breakdown frequently occurred not at the point of final reporting but during earlier analytical transitions. Lifecycle logic therefore strengthens the SOLL design by establishing that traceability must be embedded from project initiation through long-term preservation rather than appended retrospectively.

Operationally, this supports structured repositories, version control practices, documented analytical decisions, and preserved intermediate materials. Together, these elements enhance reconstructability while aligning research practice with evolving expectations for transparency within publicly supported scholarship.

FAIR Guiding Principles

The FAIR guiding principles were prioritised because they provide a clear and externally recognised definition of reuse quality. Research outputs should be findable, accessible under defined conditions, interoperable through understandable formats, and reusable with sufficient contextual metadata (Wilkinson et al., 2016).

Within the VCH context, reuse failure was frequently associated with missing documentation, ambiguous file relationships, and inconsistent storage practices rather than reluctance to share knowledge. FAIR therefore functions simultaneously as a diagnostic lens for interpreting the IS situation and as an evaluative benchmark for defining the SOLL state.

Prioritising FAIR also strengthens macro-level alignment with broader expectations for responsible research stewardship and data governance (OECD, 2021; Science Europe, 2021). Positioning the framework against recognised international standards reduces the risk that the proposed improvements appear locally constructed or procedurally idiosyncratic.

Design Science Research as Methodological Backbone

Design Science Research was confirmed as the methodological backbone of the study because the research does not aim solely to explain reproducibility breakdown. It seeks to develop and validate a structured framework capable of improving research practice.

Design science legitimises solution development as a scholarly contribution when it is grounded in theory, tested within context, and evaluated against explicit criteria (Hevner et al., 2004; Peffers et al., 2007). Retaining this perspective ensures that the framework is interpreted as a research outcome rather than as operational guidance alone.

Equally important, design science supports iterative refinement without compromising rigor. This is particularly valuable in applied laboratory environments where interventions must function under real constraints rather than idealised conditions.

By anchoring the study within design science, the transition from IS diagnosis to SOLL construction becomes methodologically coherent rather than procedurally improvised.

Socio-Technical Systems Theory

Socio-technical systems theory was prioritised because it explains why documentation breakdown rarely originates from a single cause. Organisational outcomes emerge through the interaction of social behaviours, technological infrastructures, workflow expectations, and institutional norms (Trist, 1981; Mumford, 2006).

This perspective reinforces the interpretation that traceability failure within the VCH Lab is systemic rather than behavioural. Improvements therefore cannot rely exclusively on better tools or clearer templates. Sustainable change requires alignment between supervision routines, student incentives, role clarity, and informational structures.

This insight significantly informs the SOLL orientation. A reproducible research framework must operate as a socio-technical intervention rather than a purely technical structure.

Knowledge Conversion and Organisational Continuity

Knowledge management theory, particularly the SECI model, was retained because it clarifies the mechanism through which continuity is lost. Knowledge that remains tacit is inherently fragile. When reasoning is not externalised into durable records, it becomes vulnerable to disappearance during cohort transition (Nonaka & Takeuchi, 1995; Nonaka & von Krogh, 2009).

Field evidence strongly suggested that this mechanism was active within the IS situation, as later teams often relied on verbal explanation to interpret prior work. The SECI model therefore strengthens the explanatory bridge between traceability breakdown and knowledge conversion failure.

From a design perspective, this implies that the framework must actively support externalisation and structured combination so that analytical pathways remain visible beyond the tenure of individual researchers.

Communities of Practice and Embedded Continuity

Communities of Practice theory was prioritised because long-term continuity depends on shared routines rather than isolated compliance. Communities cultivate common language, tools, and expectations that persist even as individual members change (Wenger, 1998; Wenger et al., 2002).

This perspective directly addresses the cohort turnover characteristic of the VCH Lab environment. Even a technically robust framework will not sustain itself if successive cohorts treat documentation as optional or peripheral to research competence.

Communities of Practice therefore strengthen the sustainability logic by explaining how reproducibility practices become embedded within research identity rather than perceived as administrative burden. This insight supports later design considerations related to onboarding clarity, supervisory reinforcement, shared templates, and visible norms that encourage consistent adoption.

Stage Three Outcome

Stage Three produced a defensible theoretical core capable of guiding the remainder of the study. The research can now proceed with confidence that the IS situation reflects structurally explainable mechanisms rather than isolated operational failures. The framework-oriented response is academically legitimate. Transparency and reuse can be evaluated using recognised international standards. Continuity depends on embedded practice rather than documentation alone.

Crucially, the study has now progressed beyond theoretical exploration. It operates from a position of conceptual sufficiency.

The prioritised model set provides a stable intellectual architecture for operationalisation, framework development, and evaluation within the applied laboratory environment. Because each retained perspective performs a defined analytical role, the transition toward the SOLL situation is guided by purposeful theoretical alignment rather than conceptual abundance.

At this stage, the research possesses the degree of theoretical coherence required to support design decisions without inviting critique that the framework rests on selective or underdeveloped foundations.

Conclusion: Methodological Progression and Analytical Justification

This study followed a deliberately structured research progression in which empirical investigation and theoretical development were conducted through a logically ordered and methodologically transparent sequence. The research began by establishing the IS situation, defined as the observable state of research practices within the Value Chain Hackers Lab. This diagnosis was grounded in field research designed to capture how research work was documented, transferred, and reused across student cohorts.

The field phase employed a controlled evidentiary approach. Document analysis examined whether analytical decisions, intermediate steps, data sources, and validation logic were visibly preserved within completed project documentation. Fully structured interviews with project leads and supervisory staff then clarified how research activities unfolded in practice and how decisions were recorded and communicated during project transitions. Using complementary sources enabled triangulation, strengthening interpretive credibility by ensuring that the IS situation reflected observable conditions rather than perception alone (Yin, 2018).

The evidence revealed a recurring structural vulnerability. Although projects frequently generated valuable outcomes, the analytical pathways underlying those outcomes were not consistently reconstructable. Decision rationales, transformation steps, and supporting records were often fragmented or unevenly preserved, causing later cohorts to rely on verbal explanation rather than durable documentation. When continuity depends primarily on social memory, verification becomes difficult and reuse potential declines, a pattern widely recognised within reproducibility scholarship (Peng, 2011).

Because this condition appeared across cases rather than as isolated instances, the IS situation was interpreted as a systemic weakness in research traceability and continuity rather than an individual performance issue. Sociotechnical theory supports this interpretation by demonstrating that organisational breakdown typically emerges from the interaction between routines, tools, and human actors (Trist, 1981; Mumford, 2006).

Methodological rigor was preserved through intentional research design. The interview instrument was developed using established guidance for structured qualitative inquiry, ensuring alignment between the research question, analytical constructs, and question architecture (Kallio et al., 2016). Fully structured interviews strengthened comparability across participants while maintaining analytical focus on procedural conditions (DeJonckheere & Vaughn, 2019). Construct selection was similarly grounded in prior research identifying traceability gaps, lifecycle visibility challenges, and knowledge transfer breakdown as institutional concerns rather than individual failings (Chen et al., 2022; Lalu et al., 2023; Burnette et al., 2016; Kralj & Landwehr, 2025).

An adjustment to the original research plan was required when survey distribution proved infeasible due to timing constraints. Rather than weakening the study, this adaptation strengthened procedural accuracy by prioritising participants directly involved in recently completed projects. Applied case study research recognises that rigor is preserved through transparent and justified methodological decisions that protect data quality within natural research environments (Yin, 2018).

Once the IS situation had been empirically established, the study progressed toward defining the SOLL state representing the desired future condition of research practice. This target was not constructed speculatively but informed through disciplined desk research identifying theoretical models and reproducibility standards capable of explaining the observed structural weaknesses and defining conditions for improvement.

The staged desk research process reinforced analytical legitimacy. Broad exploration confirmed that the IS situation could be explained using established theory. Independent expansion tested the stability of this explanation across multiple scholarly traditions. Selective prioritisation then ensured that the retained theoretical core directly supports diagnosis, research design legitimacy, evaluative clarity, and long term continuity.

Maintaining a traceable chain of reasoning from empirical observation to conceptual selection aligns with case study guidance on evidentiary transparency (Yin, 2018) and with design science principles that emphasise purposeful knowledge use when constructing a structured solution to a real problem (Hevner et al., 2004; Peffers et al., 2007).

Taken together, this progression demonstrates methodological orchestration rather than sequential activity. Field research established the IS situation through triangulated evidence, while theoretical development translated empirical diagnosis into stable and prioritised design requirements. The study therefore operates from a position of conceptual sufficiency.

The proposed reproducible research framework emerges not from abstract preference but from a structurally validated research need supported by convergent theory and institutional expectations for transparency, traceability, and reuse. With theoretical coherence established and the chain of evidence preserved, the research is now positioned to proceed into operationalisation and framework development with methodological confidence.

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